

WAYNE KIRK SCHMIDT

PROFILE

Principal Security Architect and Platform Engineer with 30+ years designing and building mission-critical systems for Morgan Stanley, Nomura Securities, Sumo Logic, Tanium, and global financial institutions. Hands-on experience securing AI platforms, distributed systems, cloud infrastructure, and Tier-0 production environments where security, resilience, and operational excellence are inseparable.

Designed Production Perimeter®, a patented adaptive machine learning security platform later acquired by Illumio. Built BEACON®, an ML-driven threat detection platform reducing compromise identification time by 97%; ARK®, an intelligent dependency management platform supporting more than 60,000 hosts; and MORPHOS/9STACK®, a reasoning-based AI orchestration architecture for secure enterprise AI.

Throughout my career, I have designed and built platforms that enable engineering teams to innovate with confidence by making security a foundational principle rather than a constraint, embedding governance, resilience, and secure-by-design practices into every platform.

GenAI Security · AI Platform Security · Security Engineering · Threat Modelling · Zero-Trust Architecture · AI Governance · ML-Driven Threat Detection · Distributed Systems · Kubernetes · Multi-Cloud Infrastructure

EXECUTION	LEADERSHIP	ADVISORY
Distributed, low-latency, AI-enabled systems at scale. Hands-on depth across DevOps, ML/AI, and Security from 60K+ host fleets to ULL data platforms.	Engineering and operations ownership across Tier-0 systems and \$107M+ portfolios. Teams up to 30 engineers across global and regional operations.	Supported 60+ clients across 6 industries and 4 regions, allowing SME, MSSP, and Enterprise to navigate platform risk, security, operations, and transformation.

SIGNATURE OUTCOMES

REALIZED	GUIDED	DELIVERED
\$300M+ measured impact across products, services, and portfolios in AMER and APAC.	5 zero-to-one production-grade platforms across security, finance, and operations.	90+ initiatives in system stabilization, digital transformation, and operational excellence.

- Invented **Production Perimeter®**, an adaptive ML-driven security platform deployed globally, with patented intellectual property later acquired by Illumio.
- Built **ARK®**, an ML-driven dependency intelligence platform supporting more than **60,000 hosts**, reducing deployment risk by **97%** and accelerating releases **17x**.
- Built **BEACON®**, an ML-driven threat detection platform reducing compromise identification time by 97% across enterprise endpoint fleets.
- Engineered a high-volume security event management platform processing **1.5 million** events per minute, reducing incident response from **20 minutes to 10 seconds**.
- Delivered **NXT RAW®**, a ¼µs ultra-low-latency market data platform generating **\$14M+** in revenue across multiple APAC markets.
- Designed AI-assisted operational workflows reducing Incident Management from **two person-weeks to fifteen minutes** while improving governance, consistency, and operational efficiency.

Detailed Delivery Case Studies: casestudies.changeis.io

PROFESSIONAL EXPERIENCE

ChangeIS	Founder & Principal Technologist	Tokyo, Japan	2013 – Present
Engineering practice delivering AI, security, observability, and platform engineering solutions across financial services, insurance, and technology organizations globally.			
- Engineered a Kafka-Logstash-Grafana observability platform for a US MSSP, reducing incident response time by 99.16%. - Deployed ML-driven log analysis platform reducing issue triage effort by 85% and reducing MTTR by 3x. - Created ML-driven asset platform for global insurance firm, raising accuracy to 99.9%, reducing reporting lag by 99.93%.			
Insight Innovation Fund	Head of Product (Fractional / NED)	Tokyo, Japan	2024 – Present
- Patented MORPHOS/9STACK®, a reasoning-based AI orchestration architecture for secure, private, adaptive enterprise AI deployment. - Directed product architecture and enterprise AI solutions, advising on go-to-market, partnerships, and M&A strategy.			
Sumo Logic Japan, K.K.	Senior Solution Engineer	Tokyo, Japan	2018 – 2022
- Built 10+ vendor integrations across security, observability, and DevOps, expanding platform addressability for enterprise clients. - Developed automated monitoring tooling reducing deployment time by 82%, adopted as the APAC reference tooling. - Authored 40+ client solutions across DevOps and observability, supporting renewals and expansion revenue in Japan.			

Tanium G.K. | Executive Director – Technical Account Management | Tokyo, Japan | 2015 – 2016

- Created **BEACON®**, a ML-driven compromise threat detection system, reducing compromise identification time by **97%**.
- Built **BRAVO®**, a ML-driven testing framework saving an estimated **\$1.8M** in engineering effort.
- Architected compliance reporting for **1,000+** Fuji Xerox printers via Tanium, shortening audit/reporting cycles from weeks to days.

Smart Technology Partners | Chief Technology Officer (Fractional) | Japan | 2013 – 2016

- Designed a green energy cloud architecture using refurbished hardware, delivering sustainable infrastructure for 10+ clients.
- Delivered Digital Transformation proposals for regional government clients across Fukushima and Kumamoto prefectures.
- Developed cloud solutions across logistics, genetic data processing, and data center security.

Nomura Securities | Executive Director – Asia Head of Market Data / Global Head of Market Data Engineering | Tokyo, Japan | 2009 – 2013

- Delivered **NXT RAW**, a 1/4µs ML-driven TSE data platform generating **\$14M+** in revenue.
- Built a DevOps automation framework reducing engineering effort by **70%** across a **\$25M** project portfolio.
- Managed a **\$107M** portfolio over **7 APAC** markets, **30-person** team, and **15 global vendors** supporting trading and retail functions.

Morgan Stanley Securities Japan | Vice President – Infrastructure Architect, Asia-Pacific | Tokyo, Japan | 2003 – 2009

- Built **ARK**, ML-driven dependency management across **60,000+** hosts, reducing deployment risk by **97%** and releases by **17×**.
- Invented **Production Perimeter**, a patented self-learning ML firewall preventing hundreds of millions USD in potential trading losses.
- Rebuilt Asia Equities QA and Integration Testing Environment, resolving systemic stability issues across the region.
- Grew the infrastructure development team from 3 to **12 engineers** across APAC, in addition to a global deployment team.

Morgan Stanley | Vice President – Perimeter Security Engineering | New York, New York | 1998 – 2003

- Built **AEGIS**, cross-platform security and configuration management across 80+ institutional trading locations.
- Built **ARGUS**, a firewall lifecycle automation suite coordinating global plant builds and reducing misconfiguration risk.
- Led the firm's Y2K global security infrastructure refresh, replacing at-risk legacy systems across global operations.

EARLY CAREER: BOSTON & NEW YORK

Systems and infrastructure roles establishing technical foundations in networking, distributed systems, and enterprise platforms:

- GTE Internetworking Systems: automated provisioning across **500+** servers, improving restore performance by **20×**.
- Wildfire Communications: led 4-person sysadmin team, automating Windows NT provisioning from **7 hours** to **20 minutes**.
- Pencom / Collective Technologies: architected Tivoli-based software deployment platform for a **5,000**-laptop fleet.
- Salomon Brothers: fixed income trading support, delivered **15×** reduction in risk calculation time, **\$1.3M** procurement savings.

PRODUCTS & PATENTS

MORPHOS/9STACK®	Reasoning-based AI orchestration; private, adaptive, edge-ready, with memory, context, and bounded ethics.
Production Perimeter®	ML-driven adaptive firewall; self-learning security enforcement across LAN, MAN, WAN. Acquired by Illumio.
NXT RAW® / NXT	Ultra-low latency market data delivery; sub-1/4 microsecond signal delivery at institutional scale.
BEACON®	ML-driven threat detection system; 97% reduction in compromise identification time across endpoint fleets.

Additional patents and products in event management, operational orchestration, security automation, and distributed systems architecture.

SELECTED TECHNICAL SKILLS

DevOps / Automation	Python, Bash, PowerShell, Ansible, Terraform, Puppet, Chef, Kubernetes, Docker, VMware, Jenkins, GitHub Actions, GitLab CI, AWS, GCP, Azure, OpenStack, Kafka, RabbitMQ, Tibco, Splunk, Sumo Logic, Elastic Stack, Grafana, Prometheus, Nagios
ML / AI Systems	scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, LangChain, spacy, NumPy, Pandas, Reasoning architecture design, Adaptive workflow systems, Low-latency ML inference, Edge & federated deployment, MORPHOS/9STACK® (patented), Matplotlib, Plotly, Dash
Security / Resilience	Tanium, CrowdStrike, Auth0, Recorded Future, Zeek, eBPF, Zero-trust architecture, NIST 800, ISO 27000, COBIT, SOX, ISO 31000, Checkpoint, Cisco, Arista, Palo Alto, ML-driven threat detection, Compliance & audit automation, Wireshark, Rkhunter

Additional Capabilities: waynekirkschmidt.me/technical

EDUCATION & LANGUAGES

Ohio State University — B.S. Physics (dual study: Physics & Fine Arts)

Adjunct Professor, Continuing Education — New York University · Programming, Networking, and Security

English: Native Japanese: Professional working proficiency · JLPT N3 · N2 in progress